

MariaDB_dataset-1

Commit 1:

3a43b7c60b84314135b324840b03b3e36265f072

Non-C file modifications, therefore no performance regression testing is required.

Commit 2:

66f52ba6308f1018dafd12dc90611b55845e198d

Unable to provide test cases

Commit 3:

7094a75596d7225d06b187553e38216e9b91072f

The 1st symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  55.851
4  The new version is executed by default
5  53.3547
6  The old version
7  53.4365
8  Comparison of New and old versions (along symbolic execution path)
9  0.045184471288351534
10 Comparison of New and old versions (along the default execution path)
11 -0.00153078888025977
```

No performance regression

Commit 4:

e2dbd9b6ac9573888c4a2f7ea4aef715d218b6ca

Unable to provide test cases

Commit 5:

5091986ceaca20c650d4469de08201c5b85428e4

Unable to provide test cases

Commit 6:

1fecf581acb999bac788a418ae62310fd9ca0350

Non-C file modifications, therefore no performance regression testing is required.

Commit 7:

3a81664cb80bfb1598d1a113f24a52f0a627b4f9

Non-C file modifications, therefore no performance regression testing is required.

Commit 8:

689bed19406e30621950ff2393c4b30917cbc439

Non-C file modifications, therefore no performance regression testing is required.

Commit 9:

153778437dc7dac2db469b06c3434e120b1e8198

The 1st symbolic execution

Testcase:OPTIMIZE TABLE posts;

No conditional branches, no need for symbolic execution.

代码块

- 1 end-to-end diff:
- 2 The new version executes along symbols

```
3 141.154
4 The new version is executed by default
5 141.3047
6 The old version
7 139.4807
8 Comparison of New and old versions (along symbolic execution path)
9 0.01199664182929956
10 Comparison of New and old versions (along the default execution path)
11 0.013077078047356973
```

No performance regression

The 2nd symbolic execution

Testcase:DELETE IGNORE FROM comments WHERE id = 1;

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 0.343
4 The new version is executed by default
5 0.341
6 The old version
7 0.361
8 Comparison of New and old versions (along symbolic execution path)
9 -0.0498614958
10 Comparison of New and old versions (along the default execution path)
11 -0.055401662
```

No performance regression

The 3rd symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
```

```
3 52.0379
4 The new version is executed by default
5 51.6813
6 The old version
7 55.3356
8 Comparison of New and old versions (along symbolic execution path)
9 -0.059594546729411065
10 Comparison of New and old versions (along the default execution path)
11 -0.0660388610587036
```

No performance regression

The 4th symbolic execution

Testcase:REPLACE INTO users (id, username, email) SELECT p.id, u.username, CONCAT(u.username, '@example.com') FROM users u, posts p WHERE u.id = p.user_id;

Symbolic execution exception occurred, resulting in a timeout.

old异常

No performance regression

The 5th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY u.username ORDER BY post_count DESC LIMIT 5;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 6th symbolic execution

Testcase:CREATE TABLE IF NOT EXISTS comments_new (id INT AUTO_INCREMENT PRIMARY KEY, post_id INT, comment TEXT NOT NULL, created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP, FOREIGN KEY (post_id) REFERENCES posts(id));

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 7th symbolic execution

Testcase:TRUNCATE TABLE comments;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 8th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 9th symbolic execution

Testcase:CREATE TEMPORARY TABLE IF NOT EXISTS temp_users AS SELECT * FROM users;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 10th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 11th symbolic execution

Testcase:OPTIMIZE TABLE posts;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 12th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 13th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 14th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 15th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 16th symbolic execution

Testcase:ALTER TABLE employee CHANGE eid eid INT NOT NULL;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 17th symbolic execution

Testcase:CREATE TEMPORARY TABLE IF NOT EXISTS temp_users AS SELECT * FROM users;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 18th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 19th symbolic execution

Testcase:DROP TRIGGER IF EXISTS before_user_insert;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 20th symbolic execution

Testcase:SELECT u.username, COUNT(p.id) AS post_count FROM users u, posts p WHERE u.id = p.user_id AND u.id IN (100, 200, 300, 400, 500, 600, 700, 800, 900, 1000) GROUP BY u.username ORDER BY post_count DESC LIMIT 5;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 21st symbolic execution

Testcase:SELECT * FROM employee WHERE salary > 6000 AND (department_id = 1 OR hire_date > '2025-01-01');

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 22nd symbolic execution

Testcase:TRUNCATE TABLE comments;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 23rd symbolic execution

Testcase:SELECT * FROM users WHERE id IN (SELECT * FROM (SELECT id FROM posts ORDER BY id LIMIT 2) x where id < 1000);

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 24th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 25th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 26th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 27th symbolic execution

Testcase:INSERT INTO eids(id, eid) VALUES (65536, 72004) ON DUPLICATE KEY UPDATE eid = 72004;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 28th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 29th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 30th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 31st symbolic execution

Testcase:CREATE TEMPORARY TABLE IF NOT EXISTS temp_users AS SELECT * FROM users;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 32nd symbolic execution

Testcase:OPTIMIZE TABLE posts;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 33rd symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 34th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 35th symbolic execution

Testcase:CREATE TABLE IF NOT EXISTS products_new AS SELECT * FROM products;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 36th symbolic execution

Testcase:CREATE TEMPORARY TABLE IF NOT EXISTS temp_users AS SELECT * FROM users;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 37th symbolic execution

Testcase:OPTIMIZE TABLE posts;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 38th symbolic execution

Testcase:CREATE TABLE IF NOT EXISTS comments_new (id INT AUTO_INCREMENT PRIMARY KEY, post_id INT, comment TEXT NOT NULL, created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP, FOREIGN KEY (post_id) REFERENCES posts(id));

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 39th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 40th symbolic execution

Testcase:DROP TABLE IF EXISTS posts;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 41st symbolic execution

Testcase:OPTIMIZE TABLE posts;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 42nd symbolic execution

Testcase:CREATE TABLE IF NOT EXISTS user_profiles_new (user_id INT PRIMARY KEY, bio TEXT, birthdate DATE, profile_picture CHAR(20), FOREIGN KEY (user_id) REFERENCES users(id)) ENGINE=InnoDB DEFAULT CHARSET=utf8 COLLATE=utf8_general_ci COMMENT='User Profiles Table';

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 43rd symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY u.username ORDER BY post_count DESC LIMIT 5;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 44th symbolic execution

Testcase:DROP TABLE IF EXISTS posts;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 45th symbolic execution

Testcase:INSERT INTO eids(id, eid) VALUES (65536, 72004) ON DUPLICATE KEY UPDATE eid = 72004;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 46th symbolic execution

Testcase:CREATE DATABASE IF NOT EXISTS test_db;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 47th symbolic execution

Testcase:DROP DATABASE IF EXISTS test;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 48th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 49th symbolic execution

Testcase:CREATE TEMPORARY TABLE IF NOT EXISTS temp_users AS SELECT * FROM users;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 50th symbolic execution

Testcase:ALTER TABLE employee CHANGE eid eid INT NOT NULL;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 51st symbolic execution

Testcase:TRUNCATE TABLE comments;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 52nd symbolic execution

Testcase:DELETE IGNORE FROM comments WHERE id = 1;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 53rd symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 54th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 55th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 56th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 57th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 58th symbolic execution

Testcase:CREATE TEMPORARY TABLE IF NOT EXISTS temp_users AS SELECT * FROM users;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 59th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 60th symbolic execution

Testcase:OPTIMIZE TABLE posts;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 61st symbolic execution

Testcase:DELETE FROM users WHERE NOT id < 1000;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 62nd symbolic execution

Testcase:CREATE OR REPLACE VIEW user_post_comments_new AS SELECT u.username, p.title, c.comment FROM users u JOIN posts p ON u.id = p.user_id JOIN comments c ON p.id = c.post_id;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

Commit 10:

717c12de0e2dcf9118777c28eeae3b766e0886f1

The 1st symbolic execution

Testcase:ALTER TABLE employee CHANGE eid eid INT NOT NULL;

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
```

No performance regression

The 2nd symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  5.2941
4  The new version is executed by default
5  5.3477
6  The old version
7  5.9139
8  Comparison of New and old versions (along symbolic execution path)
9  -0.10480393648861153
10 Comparison of New and old versions (along the default execution path)
11 -0.09574054346539514
```

No performance regression

The 3rd symbolic execution

Testcase:OPTIMIZE TABLE posts;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
```

No performance regression

The 4th symbolic execution

Testcase:SELECT DISTINCT id, username, (SELECT COUNT(*) FROM posts WHERE user_id = users.id) * 2 AS post_count_double FROM users WHERE (SELECT COUNT(*) FROM posts WHERE user_id = users.id) * 2 > 4;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  149.025500000000002
4  The new version is executed by default
5  155.4743
6  The old version
7  155.1432
8  Comparison of New and old versions (along symbolic execution path)
9  -0.039432601622243095
10 Comparison of New and old versions (along the default execution path)
11 0.002134157346245225
```

No performance regression

The 5th symbolic execution

Testcase:SELECT u.username, u.email FROM users u JOIN (SELECT user_id FROM posts ORDER BY created_at DESC LIMIT 5) p ON u.id = p.user_id;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.0764
4  The new version is executed by default
5  0.0868
6  The old version
7  0.0856
8  Comparison of New and old versions (along symbolic execution path)
9  -0.10747663551401869
10 Comparison of New and old versions (along the default execution path)
11 0.014018691588785125
```

No performance regression

The 6th symbolic execution

Testcase:DELETE IGNORE FROM comments WHERE id = 1;

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
```

No performance regression

The 7th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 0.337400000000000003
4 The new version is executed by default
5 0.3288
6 The old version
7 0.3361
8 Comparison of New and old versions (along symbolic execution path)
9 0.0038678964593871564
10 Comparison of New and old versions (along the default execution path)
11 -0.02171972627194296
```

No performance regression

The 8th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND u.id IN (SELECT user_id FROM posts WHERE created_at > '2025-01-
01') GROUP BY u.username HAVING COUNT(p.id) > 1;

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
```

```
2 The new version executes along symbols
3 0.7902
4 The new version is executed by default
5 0.7804
6 The old version
7 0.7689
8 Comparison of New and old versions (along symbolic execution path)
9 0.027701911822083477
10 Comparison of New and old versions (along the default execution path)
11 0.01495643126544408
```

No performance regression

The 9th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 0.3517
4 The new version is executed by default
5 0.3289
6 The old version
7 0.3351
8 Comparison of New and old versions (along symbolic execution path)
9 0.049537451507012845
10 Comparison of New and old versions (along the default execution path)
11 -0.01850193971948667
```

No performance regression

The 10th symbolic execution

Testcase:SELECT * FROM employee WHERE department_id = 1 ORDER BY salary LIMIT 10;

No conditional branches, no need for symbolic execution.

代码块
end-to-end diff:

```
2 The new version executes along symbols
3 0.0878
4 The new version is executed by default
5 0.0877
6 The old version
7 0.0875
8 Comparison of New and old versions (along symbolic execution path)
9 0.00342857143
10 Comparison of New and old versions (along the default execution path)
11 0.00228571429
```

No performance regression

The 11th symbolic execution

Testcase:SELECT department_id, AVG(salary) AS avg_salary FROM employee WHERE department_id = 1 GROUP BY salary ORDER BY department_id;

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 0.0354
4 The new version is executed by default
5 0.02750000000000000004
6 The old version
7 0.0356
8 Comparison of New and old versions (along symbolic execution path)
9 -0.005617977528089854
10 Comparison of New and old versions (along the default execution path)
11 -0.22752808988764034
```

No performance regression

The 12th symbolic execution

Testcase:SHOW DATABASES;

No conditional branches, no need for symbolic execution.

代码块
end-to-end diff:

```
2 The new version executes along symbols
3 0.0629
4 The new version is executed by default
5 0.0627
6 The old version
7 0.0624
8 Comparison of New and old versions (along symbolic execution path)
9 0.00801282051
10 Comparison of New and old versions (along the default execution path)
11 0.00480769231
```

No performance regression

The 13th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 5.336399999999999
4 The new version is executed by default
5 5.5334
6 The old version
7 5.479
8 Comparison of New and old versions (along symbolic execution path)
9 -0.026026647198394
10 Comparison of New and old versions (along the default execution path)
11 0.009928819127578066
```

No performance regression

The 14th symbolic execution

Testcase:SELECT * FROM users WHERE id IN (SELECT * FROM (SELECT id FROM posts ORDER BY id LIMIT 2) x where id < 1000);

No conditional branches, no need for symbolic execution.

代码块
end-to-end diff:

```
2 The new version executes along symbols
3 0.0078
4 The new version is executed by default
5 0.0078
6 The old version
7 0.0077
8 Comparison of New and old versions (along symbolic execution path)
9 0.012987012987012908
10 Comparison of New and old versions (along the default execution path)
11 0.012987012987012908
```

No performance regression

The 15th symbolic execution

Testcase:SELECT * FROM users WHERE id IN (SELECT * FROM (SELECT id FROM posts ORDER BY id LIMIT 2) x where id < 1000);

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 0.0057
4 The new version is executed by default
5 0.0077
6 The old version
7 0.0078
8 Comparison of New and old versions (along symbolic execution path)
9 -0.26923076923076916
10 Comparison of New and old versions (along the default execution path)
11 -0.012820512820512744
```

No performance regression

The 16th symbolic execution

Testcase:SELECT * FROM (SELECT user_id FROM posts UNION SELECT user_id FROM comments) AS derived_table WHERE user_id > 10;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.4415
4  The new version is executed by default
5  0.4404
6  The old version
7  0.4456000000000000005
8  Comparison of New and old versions (along symbolic execution path)
9  -0.009201077199281973
10 Comparison of New and old versions (along the default execution path)
11 -0.01166965888689416
```

No performance regression

The 17th symbolic execution

Testcase:CALL GetUserPosts(1);

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.0002
4  The new version is executed by default
5  0.0002
6  The old version
7  0.0002
8  Comparison of New and old versions (along symbolic execution path)
9  0.0
10 Comparison of New and old versions (along the default execution path)
11 0.0
```

No performance regression

The 18th symbolic execution

Testcase:SELECT * FROM user_post_comments;

No conditional branches, no need for symbolic execution.

代码块
1 end-to-end diff:

```
2 The new version executes along symbols
3 1.166200000000000001
4 The new version is executed by default
5 1.1669
6 The old version
7 1.1775
8 Comparison of New and old versions (along symbolic execution path)
9 -0.009596602972399037
10 Comparison of New and old versions (along the default execution path)
11 -0.009002123142250483
```

No performance regression

The 19th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 5.1761
4 The new version is executed by default
5 5.1574
6 The old version
7 5.3256999999999999
8 Comparison of New and old versions (along symbolic execution path)
9 -0.028090204104624657
10 Comparison of New and old versions (along the default execution path)
11 -0.03160147961770274
```

No performance regression

The 20th symbolic execution

Testcase:SELECT department_id, AVG(salary) AS avg_salary FROM employee WHERE department_id = 1 GROUP BY department_id ORDER BY department_id;

No conditional branches, no need for symbolic execution.

代码块
1 end-to-end diff:

```
2 The new version executes along symbols
3 0.0198
4 The new version is executed by default
5 0.0278000000000000002
6 The old version
7 0.0295
8 Comparison of New and old versions (along symbolic execution path)
9 -0.3288135593220338
10 Comparison of New and old versions (along the default execution path)
11 -0.057627118644067686
```

No performance regression

The 21st symbolic execution

Testcase:SELECT username FROM users WHERE id BETWEEN 700 AND 1200 ORDER BY username;

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 0.0305
4 The new version is executed by default
5 0.0422
6 The old version
7 0.043
8 Comparison of New and old versions (along symbolic execution path)
9 -0.2906976744186046
10 Comparison of New and old versions (along the default execution path)
11 -0.018604651162790586
```

No performance regression

The 22nd symbolic execution

Testcase:DESC users;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.0154
4  The new version is executed by default
5  0.0154
6  The old version
7  0.0214
8  Comparison of New and old versions (along symbolic execution path)
9  -0.28037383177570085
10 Comparison of New and old versions (along the default execution path)
11 -0.28037383177570085
```

No performance regression

The 23rd symbolic execution

Testcase:SELECT username FROM users WHERE id = 1000;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.0038
4  The new version is executed by default
5  0.0036999999999999997
6  The old version
7  0.0038
8  Comparison of New and old versions (along symbolic execution path)
9  0.0
10 Comparison of New and old versions (along the default execution path)
11 -0.02631578947368428
```

No performance regression

The 24th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.3357
4  The new version is executed by default
5  0.3332
6  The old version
7  0.34199999999999997
8  Comparison of New and old versions (along symbolic execution path)
9  -0.01842105263157887
10 Comparison of New and old versions (along the default execution path)
11 -0.025730994152046712
```

No performance regression

The 25th symbolic execution

Testcase:SHOW DATABASES;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.0055
4  The new version is executed by default
5  0.0056
6  The old version
7  0.0056
8  Comparison of New and old versions (along symbolic execution path)
9  -0.017857142857142905
10 Comparison of New and old versions (along the default execution path)
11 0.0
```

No performance regression

The 26th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

No conditional branches, no need for symbolic execution.

代码块
1 end-to-end diff:

```
2 The new version executes along symbols
3 5.1894
4 The new version is executed by default
5 5.4955
6 The old version
7 5.102
8 Comparison of New and old versions (along symbolic execution path)
9 0.017130537044296294
10 Comparison of New and old versions (along the default execution path)
11 0.07712661701293601
```

No performance regression

The 27th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND u.id IN (SELECT user_id FROM posts WHERE created_at > '2025-01-01') GROUP BY u.username HAVING COUNT(p.id) > 1;

No conditional branches, no need for symbolic execution.

代码块

```
1 end-to-end diff:
2 The new version executes along symbols
3 0.7828
4 The new version is executed by default
5 0.78470000000000001
6 The old version
7 0.7823
8 Comparison of New and old versions (along symbolic execution path)
9 0.0006391409945034589
10 Comparison of New and old versions (along the default execution path)
11 0.003067876773616348
```

No performance regression

The 28th symbolic execution

Testcase:SELECT DISTINCT id, username, (SELECT COUNT(*) FROM posts WHERE user_id =
users.id) * 2 AS post_count_double FROM users WHERE (SELECT COUNT(*) FROM posts WHERE
user_id = users.id) * 2 > 4;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  149.3683
4  The new version is executed by default
5  155.0746
6  The old version
7  155.055400000000002
8  Comparison of New and old versions (along symbolic execution path)
9  -0.03667785836546173
10 Comparison of New and old versions (along the default execution path)
11 0.00012382670967914483
```

No performance regression

The 29th symbolic execution

Testcase:REPLACE INTO users (username, email) SELECT username, CONCAT(username, '@example.com') FROM users WHERE id < 1000;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  4.2323
4  The new version is executed by default
5  4.2004
6  The old version
7  4.2630
8  Comparison of New and old versions (along symbolic execution path)
9  -0.00720150129
10 Comparison of New and old versions (along the default execution path)
11 -0.0146844945
```

No performance regression

The 30th symbolic execution

Testcase:INSERT INTO users (id, username, email) SELECT id, username, CONCAT(username, '@example.com') FROM users WHERE id < 1000 ON DUPLICATE KEY UPDATE email = VALUES(email);

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.3443
4  The new version is executed by default
5  0.3479
6  The old version
7  0.3458
8  Comparison of New and old versions (along symbolic execution path)
9  -0.004337767495662236
10 Comparison of New and old versions (along the default execution path)
11 0.006072874493927099
```

No performance regression

The 31st symbolic execution

Testcase:DELETE FROM comments WHERE id IN (SELECT c.id FROM (SELECT comments.id FROM comments JOIN posts ON comments.post_id = posts.id WHERE posts.user_id = 1 ORDER BY comments.created_at DESC LIMIT 5) AS c);

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.9648999999999999
4  The new version is executed by default
5  0.9624
6  The old version
7  0.9821
8  Comparison of New and old versions (along symbolic execution path)
9  -0.01751349149781092
10 Comparison of New and old versions (along the default execution path)
11 -0.020059057122492556
```

No performance regression

The 32nd symbolic execution

Testcase:DELETE FROM comments WHERE id IN (SELECT c.id FROM (SELECT comments.id FROM comments JOIN posts ON comments.post_id = posts.id WHERE posts.user_id = 1 ORDER BY comments.created_at DESC LIMIT 5) AS c);

No conditional branches, no need for symbolic execution.

No performance regression

The 33rd symbolic execution

Testcase:SELECT username, email, ROW_NUMBER() OVER (PARTITION BY username ORDER BY created_at DESC) AS row_num FROM users LIMIT 5;

No conditional branches, no need for symbolic execution.

No performance regression

The 34th symbolic execution

Testcase:WITH recent_posts AS (SELECT * FROM posts WHERE created_at > '2025-01-01')
SELECT u.username, p.title FROM users u JOIN recent_posts p ON u.id = p.user_id;

No conditional branches, no need for symbolic execution.

No performance regression

The 35th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

No conditional branches, no need for symbolic execution.

No performance regression

The 36th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

No performance regression

The 37th symbolic execution

Testcase:CREATE OR REPLACE VIEW user_post_comments_new AS SELECT u.username, p.title, c.comment FROM users u JOIN posts p ON u.id = p.user_id JOIN comments c ON p.id = c.post_id;

No conditional branches, no need for symbolic execution.

No performance regression

The 38th symbolic execution

Testcase:SELECT u.username, COUNT(p.id) AS post_count, MAX(p.created_at) AS last_post_time FROM users u LEFT JOIN posts p ON u.id = p.user_id GROUP BY u.username HAVING COUNT(p.id) > 1 AND MAX(p.created_at) > '2025-01-01';

No conditional branches, no need for symbolic execution.

No performance regression

The 39th symbolic execution

Testcase:CREATE OR REPLACE VIEW user_post_comments_new AS SELECT u.username, p.title, c.comment FROM users u JOIN posts p ON u.id = p.user_id JOIN comments c ON p.id = c.post_id;

No conditional branches, no need for symbolic execution.

No performance regression

The 40th symbolic execution

Testcase:SELECT * FROM products WHERE FIND_IN_SET('New', tags);

No conditional branches, no need for symbolic execution.

No performance regression

The 41st symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

No performance regression

The 42nd symbolic execution

Testcase:UPDATE employee SET age = age + 1 WHERE age < 30;

No conditional branches, no need for symbolic execution.

No performance regression

The 43rd symbolic execution

Testcase:ALTER TABLE employee CHANGE eid eid INT NOT NULL;

No conditional branches, no need for symbolic execution.

No performance regression

The 44th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

No performance regression

The 45th symbolic execution

Testcase:SELECT * FROM products WHERE category = 'Electronics';

No conditional branches, no need for symbolic execution.

No performance regression

The 46th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p
WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY
u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

No performance regression

The 47th symbolic execution

Testcase:SELECT * FROM employee WHERE age != 30;

No conditional branches, no need for symbolic execution.

No performance regression

The 48th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

No conditional branches, no need for symbolic execution.

No performance regression

The 49th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count, AVG(LENGTH(p.content))
AS avg_content_length FROM users u LEFT JOIN posts p ON u.id = p.user_id GROUP BY
u.username HAVING COUNT(p.id) > 1 AND AVG(LENGTH(p.content)) > 100;

No conditional branches, no need for symbolic execution.

No performance regression

The 50th symbolic execution

Testcase:SELECT u.username, u.email FROM users u WHERE EXISTS (SELECT 1 FROM posts p
WHERE p.user_id = u.id AND p.created_at > '2025-01-01') AND u.email LIKE '%example.com';

No conditional branches, no need for symbolic execution.

No performance regression

The 51st symbolic execution

Testcase:SELECT * FROM employee WHERE department_id = 1 AND salary > 6000;

No conditional branches, no need for symbolic execution.

No performance regression

The 52nd symbolic execution

Testcase:SELECT * FROM employee WHERE department_id = 1 AND salary > 6000;

No conditional branches, no need for symbolic execution.

No performance regression

The 53rd symbolic execution

Testcase:SELECT u.username, COUNT(p.id) AS post_count FROM users u, posts p WHERE u.id = p.user_id AND u.id IN (100, 200, 300, 400, 500, 600, 700, 800, 900, 1000) GROUP BY u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

No performance regression

The 54th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p WHERE u.id = p.user_id AND u.id IN (SELECT user_id FROM posts WHERE created_at > '2025-01-01') GROUP BY u.username HAVING COUNT(p.id) > 1;

No conditional branches, no need for symbolic execution.

No performance regression

The 55th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

No performance regression

The 56th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

No conditional branches, no need for symbolic execution.

No performance regression

The 57th symbolic execution

Testcase:SELECT u.username, COUNT(p.id) AS post_count FROM users u, posts p WHERE u.id = p.user_id AND u.id IN (100, 200, 300, 400, 500, 600, 700, 800, 900, 1000) GROUP BY u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

No performance regression

The 58th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p WHERE u.id = p.user_id AND u.id IN (SELECT user_id FROM posts WHERE created_at > '2025-01-01') GROUP BY u.username HAVING COUNT(p.id) > 1;

No conditional branches, no need for symbolic execution.

No performance regression

The 59th symbolic execution

Testcase:WITH recent_posts AS (SELECT * FROM posts WHERE created_at > '2025-01-01')
SELECT u.username, p.title FROM users u JOIN recent_posts p ON u.id = p.user_id;

No conditional branches, no need for symbolic execution.

No performance regression

The 60th symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

No conditional branches, no need for symbolic execution.

No performance regression

The 61st symbolic execution

Testcase:SELECT DISTINCT id, username, (SELECT COUNT(*) FROM posts WHERE user_id = users.id) * 2 AS post_count_double FROM users WHERE (SELECT COUNT(*) FROM posts WHERE user_id = users.id) * 2 > 4;

No conditional branches, no need for symbolic execution.

No performance regression

The 62nd symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

No performance regression

The 63rd symbolic execution

Testcase:UPDATE comments SET post_id=post_id+1 WHERE id>1000;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 64th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY u.username ORDER BY post_count DESC LIMIT 5;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 65th symbolic execution

Testcase:SELECT username FROM users WHERE id BETWEEN 700 AND 1200;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 66th symbolic execution

Testcase:SELECT username FROM users WHERE id BETWEEN 700 AND 1200;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

The 67th symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY u.username ORDER BY post_count DESC LIMIT 5;

Symbolic execution exception occurred, resulting in a timeout.

No performance regression

Commit 11:

feb1cf9086e7a3d9ddde78a48ada801ee9913285

Unable to provide test cases

Commit 12:

543ebbcf8e07b3c84e7a920ffa597aa81f3655b6

Unable to provide test cases

Commit 13:

c3f21762e9db30c4a5dd1e9ac676dbdafa303d4a

Unable to provide test cases

Commit 14:

261d5520a2ec003045f83d2518623dedafe2ea4c

Non-C file modifications, therefore no performance regression testing is required.

Commit 15:

7e4233746e28cf33a24d6d61e31b82f9838d635d

The 1st symbolic execution

Testcase:SELECT DISTINCT u.username, COUNT(p.id) AS post_count FROM users u, posts p WHERE u.id = p.user_id AND p.created_at BETWEEN '2025-01-01' AND '2025-12-31' GROUP BY u.username ORDER BY post_count DESC LIMIT 5;

No conditional branches, no need for symbolic execution.

代码块

```
1  end-to-end diff:
2  The new version executes along symbols
3  0.3382
4  The new version is executed by default
5  0.335300000000000004
6  The old version
7  0.3593
8  Comparison of New and old versions (along symbolic execution path)
9  -0.05872529919287506
10 Comparison of New and old versions (along the default execution path)
11 -0.06679654884497625
```

No performance regression

Commit 16:

24b5c3021dd4c3331092d32849b074b1370e6c58

Non-C file modifications, therefore no performance regression testing is required.

Commit 17:

923094b4cd58478befd31932a312ba9e4a8262f7

Non-C file modifications, therefore no performance regression testing is required.

Commit 18:

31adb3030ce531009457913cbf947456ad606646

Non-C file modifications, therefore no performance regression testing is required.

Commit 19:

c34bb80b3dd99452a9fe2a8fcfada0b714fa46fa

Non-C file modifications, therefore no performance regression testing is required.

Commit 20:

5f7e88333682877ad0368b53575511ab02c7365a

Unable to provide test cases